12 Blockchain: The road ahead

This chapter covers

- Exploring decentralized identity management
- Understanding consensus among decentralized participants
- Reviewing scalability, privacy, security, and confidentiality
- Analyzing public, private and permissioned blockchain networks
- Capturing the scientific research behind blockchain concepts

Any emerging technology will experience challenges as it is maturing. Blockchain is no exception. This field is churning with activities and initiatives in a quest for continuous improvement in technology. Although blockchain is a brilliant technology for trusted transactions, social interaction, and commerce, it is also open and decentralized. Openness and inclusivity of decentralized participants are two of the many impediments to the ready adoption of the technology. Dapps deployed on the blockchain address these concerns by enabling trusted transactions. Now that you are armed with the knowledge from earlier chapters, I encourage you to examine the challenges at all levels of the blockchain stack, from decentralized application development to contribution to protocol improvements.

In this chapter, you'll learn about some nonfunctional attributes that are relevant to blockchain applications. You need to pay